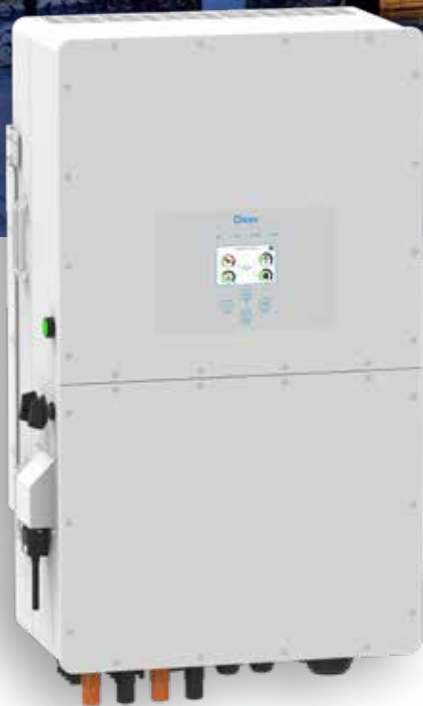


# Three Phase Hybrid Inverter

SUN-29.9/30/35K-SG01HP3-EU-BM3

SUN-40/50K-SG01HP3-EU-BM4



- 100** 100% unbalanced output, each phase; Max. output up to 50% rated power
-  DC couple and AC couple to retrofit existing solar system
- 10** \*Max. 10 pcs parallel for on-grid and off-grid operation; Support multiple batteries parallel
- 100** Max. charging/discharging current of 100A
- H** High voltage battery, higher efficiency
- 6** 6 time periods for battery charging/discharging
-  Support storing energy from diesel generator

**Deye**

Stock Code: 605117.SH

| Model                                             | SUN-29.9K-SG01HP3<br>-EU-BM3                                                                                                                                                                                         | SUN-30K-SG01HP3<br>-EU-BM3 | SUN-35K-SG01HP3<br>-EU-BM3 | SUN-40K-SG01HP3<br>-EU-BM4 | SUN-50K-SG01HP3<br>-EU-BM4 |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------|----------------------------|----------------------------|
| Battery Input Data                                |                                                                                                                                                                                                                      |                            |                            |                            |                            |
| Battery Type                                      | Lithium-ion                                                                                                                                                                                                          |                            |                            |                            |                            |
| Battery Voltage Range (V)                         | 160-800                                                                                                                                                                                                              |                            |                            |                            |                            |
| Max. Charging Current (A)                         | 50+50                                                                                                                                                                                                                |                            |                            |                            |                            |
| Max. Discharging Current (A)                      | 50+50                                                                                                                                                                                                                |                            |                            |                            |                            |
| Number of Battery Input                           | 2                                                                                                                                                                                                                    |                            |                            |                            |                            |
| Charging Strategy for Li-Ion Battery              | Self-adaption to BMS                                                                                                                                                                                                 |                            |                            |                            |                            |
| PV String Input Data                              |                                                                                                                                                                                                                      |                            |                            |                            |                            |
| Max. DC Input Power (W)                           | 38870                                                                                                                                                                                                                | 39000                      | 45500                      | 52000                      | 65000                      |
| Max. DC Input Voltage (V)                         | 1000                                                                                                                                                                                                                 |                            |                            |                            |                            |
| Start-up Voltage (V)                              | 180                                                                                                                                                                                                                  |                            |                            |                            |                            |
| MPPT Range (V)                                    | 150-850                                                                                                                                                                                                              |                            |                            |                            |                            |
| Full Load DC Voltage Range (V)                    | 360-850                                                                                                                                                                                                              | 360-850                    | 360-850                    | 360-850                    | 450-850                    |
| Rated DC Input Voltage (V)                        | 600                                                                                                                                                                                                                  |                            |                            |                            |                            |
| PV Input Current (A)                              | 36+36+36                                                                                                                                                                                                             | 36+36+36                   | 36+36+36                   | 36+36+36+36                |                            |
| Max. PV I <sub>SC</sub> (A)                       | 55+55+55                                                                                                                                                                                                             | 55+55+55                   | 55+55+55                   | 55+55+55+55                |                            |
| No.of MPP Trackers                                | 3                                                                                                                                                                                                                    |                            |                            | 4                          |                            |
| No.of Strings per MPP Tracker                     | 2+2+2                                                                                                                                                                                                                |                            |                            | 2+2+2+2                    |                            |
| AC Output Data                                    |                                                                                                                                                                                                                      |                            |                            |                            |                            |
| Rated AC Output Active Power (W)                  | 29900                                                                                                                                                                                                                | 30000                      | 35000                      | 40000                      | 50000                      |
| Max AC Output Active Power (W)                    | 29900                                                                                                                                                                                                                | 33000                      | 38500                      | 44000                      | 55000                      |
| AC Output Rated Current (A)                       | 45.4/43.4                                                                                                                                                                                                            | 45.5/43.5                  | 53.1/50.8                  | 60.7/58                    | 75.8/72.5                  |
| Max. AC Output Rated Current (A)                  | 45.4/43.4                                                                                                                                                                                                            | 50/47.8                    | 58.4/55.8                  | 66.7/63.8                  | 83.4/79.7                  |
| Max. Three-phase Unbalanced Output Current (A)    | 60                                                                                                                                                                                                                   | 60                         | 60                         | 70                         | 83.3                       |
| Max. Continuous AC Passthrough (A)                | 200                                                                                                                                                                                                                  |                            |                            |                            |                            |
| Peak Power (Off Grid)                             | 1.5 time of rated power, 10 S                                                                                                                                                                                        |                            |                            |                            |                            |
| Generator Input/Smart Load /AC Couple Current (A) | 45.4 / 200 / 45.4                                                                                                                                                                                                    | 45.5 / 200 / 45.5          | 53.1 / 200 / 53.1          | 60.7 / 200 / 60.7          | 75.8 / 200 / 75.8          |
| Power Factor Adjustment Range                     | 0.8 leading to 0.8 lagging                                                                                                                                                                                           |                            |                            |                            |                            |
| Output Frequency and Voltage                      | 50/60Hz; 3L/N/PE 220/380, 230/400Vac                                                                                                                                                                                 |                            |                            |                            |                            |
| Grid Type                                         | Three Phase                                                                                                                                                                                                          |                            |                            |                            |                            |
| Total Harmonic Distortion (THD)                   | <3% (of nominal power)                                                                                                                                                                                               |                            |                            |                            |                            |
| DC Current Injection                              | <0.5% In                                                                                                                                                                                                             |                            |                            |                            |                            |
| Efficiency                                        |                                                                                                                                                                                                                      |                            |                            |                            |                            |
| Max. Efficiency                                   | 97.60%                                                                                                                                                                                                               |                            |                            |                            |                            |
| Euro Efficiency                                   | 97.00%                                                                                                                                                                                                               |                            |                            |                            |                            |
| MPPT Efficiency                                   | 99.90%                                                                                                                                                                                                               |                            |                            |                            |                            |
| Protection                                        |                                                                                                                                                                                                                      |                            |                            |                            |                            |
| Integrated                                        | Anti-islanding Protection, PV String Input Reverse Polarity Protection, Insulation Resistor Detection, Residual Current Monitoring Unit, Output Over Current Protection, Output Shorted Protection, Surge Protection |                            |                            |                            |                            |
| Over Voltage Category                             | DC Type II/AC Type III                                                                                                                                                                                               |                            |                            |                            |                            |
| Certifications and Standards                      |                                                                                                                                                                                                                      |                            |                            |                            |                            |
| Grid Regulation                                   | VDE4105, IEC61727/62116, VDE0126, AS4777.2, CEI 0 21, EN50549-1, G98, G99, C10-11, UNE217002, NBR16149/NBR16150                                                                                                      |                            |                            |                            |                            |
| Safety EMC / Standard                             | IEC/EN 61000-6-1/2/3/4, IEC/EN 62109-1, IEC/EN 62109-2                                                                                                                                                               |                            |                            |                            |                            |
| General Data                                      |                                                                                                                                                                                                                      |                            |                            |                            |                            |
| Operating Temperature Range (°C)                  | -40-60°C, >45°C Derating                                                                                                                                                                                             |                            |                            |                            |                            |
| Cooling                                           | Smart Cooling                                                                                                                                                                                                        |                            |                            |                            |                            |
| Noise (dB)                                        | ≤65 dB                                                                                                                                                                                                               |                            |                            |                            |                            |
| Communication with BMS                            | RS485; CAN                                                                                                                                                                                                           |                            |                            |                            |                            |
| Weight (kg)                                       | 80                                                                                                                                                                                                                   |                            |                            |                            |                            |
| Cabinet Size (WxHxD mm)                           | 527×894×294 (Excluding Connectors and Brackets)                                                                                                                                                                      |                            |                            |                            |                            |
| Protection Degree                                 | IP65                                                                                                                                                                                                                 |                            |                            |                            |                            |
| Installation Style                                | Wall-mounted                                                                                                                                                                                                         |                            |                            |                            |                            |
| Warranty                                          | 5 Years (10 Years Optional)                                                                                                                                                                                          |                            |                            |                            |                            |

\*Note: The function of Multiple units work in parallel mode will be available in Q1 2023